

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

TIRE PRESSURE MONITORING SYSTEM CONTROL MODULE (TPMSCM)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable.
Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Tire Pressure Monitoring System Control Module (TPMSCM). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

For additional information, refer to: [Tire Pressure Monitoring System](#) (204-

04 Wheels and Tires, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1182-17	Tire Pressure Monitoring System - Circuit voltage above threshold	Battery/charging system fault	Refer to the relevant section of the workshop manual and test the battery and charging system
B1182-96	Tire Pressure Monitoring System - Component internal failure	Tire pressure monitoring system control module internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new tire pressure monitoring system control module
C1A56-4A	Left Front Tire Pressure Sensor and Transmitter Assembly - Incorrect component installed	Incorrect front left tire pressure sensor installed	NOTE: There will be no warning indicator/message displayed unless at least two incorrect tire pressure sensors have been installed - Using the manufacturer approved diagnostic system, perform routine - Tire Pressure Monitoring - Tire Pressure Sensor Test - If the front left tire pressure sensor fails, install a new front left tire pressure sensor
C1A56-93	Left Front Tire Pressure Sensor and Transmitter Assembly - No operation	- Incorrect wheel/tire installed (without tire pressure sensor - Winter wheels, spare wheel, etc) - Front left tire pressure sensor damaged - Front left tire pressure sensor	NOTE: Tire pressure sensor wheel position information may be incorrect if the wheels have been swapped during or after the failure event Check that the wheels and tires comply with the manufacturer's specification for the vehicle

		internal failure or signal corrupted by RF interference	▪ Check the front left tire pressure sensor for damage. Install a new front left tire pressure sensor as necessary ▪ Using the manufacturer approved diagnostic system, perform routine - Tire Pressure Monitoring - Tire Pressure Sensor Test ▪ If all tire pressure sensors pass, clear the DTCs and road test the vehicle ▪ If all tire pressure sensors fail, check for sources of RF interference (electronic devices, chargers, non-genuine accessories, etc) ▪ If some tire pressure sensors fail, install new tire pressure sensors as necessary
C1A58-4A	Right Front Tire Pressure Sensor and Transmitter Assembly - Incorrect component installed	▪ Incorrect front right tire pressure sensor installed	NOTE: There will be no warning indicator/message displayed unless at least two incorrect tire pressure sensors have been installed ▪ Using the manufacturer approved diagnostic system, perform routine - Tire Pressure Monitoring - Tire Pressure Sensor Test ▪ If the front right tire pressure sensor fails, install a new front right tire pressure sensor
C1A58-93	Right Front Tire Pressure Sensor and Transmitter Assembly - No operation	▪ Incorrect wheel/tire installed (without tire pressure sensor - Winter wheels, spare wheel, etc) ▪ Front right tire pressure sensor damaged	NOTE: Tire pressure sensor wheel position information may be incorrect if the wheels have been swapped during or after the failure event ▪ Check that the wheels and tires

		Front right tire pressure sensor internal failure or signal corrupted by RF interference	comply with the manufacturer's specification for the vehicle - Check the front right tire pressure sensor for damage. Install a new front right tire pressure sensor as necessary - Using the manufacturer approved diagnostic system, perform routine - Tire Pressure Monitoring - Tire Pressure Sensor Test - If all tire pressure sensors pass, clear the DTCs and road test the vehicle - If all tire pressure sensors fail, check for sources of RF interference (electronic devices, chargers, non-genuine accessories, etc) - If some tire pressure sensors fail, install new tire pressure sensors as necessary
C1A60-4A	Left Rear Tire Pressure Sensor and Transmitter Assembly - Incorrect component installed	Incorrect rear left tire pressure sensor installed	NOTE: There will be no warning indicator/message displayed unless at least two incorrect tire pressure sensors have been installed - Using the manufacturer approved diagnostic system, perform routine - Tire Pressure Monitoring - Tire Pressure Sensor Test - If the rear left tire pressure sensor fails, install a new rear left tire pressure sensor
C1A60-93	Left Rear Tire Pressure Sensor and Transmitter Assembly - No operation	Incorrect wheel/tire installed (without tire pressure sensor - Winter wheels, spare wheel, etc)	

		▪ Rear left tire pressure sensor damaged ▪ Rear left tire pressure sensor internal failure or signal corrupted by RF interference	NOTE: Tire pressure sensor wheel position information may be incorrect if the wheels have been swapped during or after the failure event ▪ Check that the wheels and tires comply with the manufacturer's specification for the vehicle ▪ Check the rear left tire pressure sensor for damage. Install a new rear left tire pressure sensor as necessary ▪ Using the manufacturer approved diagnostic system, perform routine - Tire Pressure Monitoring - Tire Pressure Sensor Test ▪ If all tire pressure sensors pass, clear the DTCs and road test the vehicle ▪ If all tire pressure sensors fail, check for sources of RF interference (electronic devices, chargers, non-genuine accessories, etc) ▪ If some tire pressure sensors fail, install new tire pressure sensors as necessary
C1A62-4A	Right Rear Tire Pressure Sensor and Transmitter Assembly - Incorrect component installed	▪ Incorrect rear right tire pressure sensor installed	NOTE: There will be no warning indicator/message displayed unless at least two incorrect tire pressure sensors have been installed ▪ Using the manufacturer approved diagnostic system, perform routine - Tire Pressure Monitoring - Tire Pressure Sensor Test ▪ If the rear right tire pressure sensor fails, install a new rear

			right tire pressure sensor
C1A62-93	Right Rear Tire Pressure Sensor and Transmitter Assembly - No operation	- Incorrect wheel/tire installed (without tire pressure sensor - Winter wheels, spare wheel, etc) - Rear right tire pressure sensor damaged - Rear right tire pressure sensor internal failure or signal corrupted by RF interference	NOTE: Tire pressure sensor wheel position information may be incorrect if the wheels have been swapped during or after the failure event - Check that the wheels and tires comply with the manufacturer's specification for the vehicle - Check the rear right tire pressure sensor for damage. Install a new rear right tire pressure sensor as necessary - Using the manufacturer approved diagnostic system, perform routine - Tire Pressure Monitoring - Tire Pressure Sensor Test - If all tire pressure sensors pass, clear the DTCs and road test the vehicle - If all tire pressure sensors fail, check for sources of RF interference (electronic devices, chargers, non-genuine accessories, etc) - If some tire pressure sensors fail, install new tire pressure sensors as necessary
C1D18-00	Wheel Localisation Failed - No sub type information	- One or more incorrect tire pressure sensors installed - Missing/invalid data from the anti-lock brake system control module	NOTE: There will be no warning indicator/message displayed unless at least two incorrect tire pressure sensors have been installed Using the manufacturer approved diagnostic system, check for tire pressure sensor related DTCs and

			perform the relevant corrective actions Using the manufacturer approved diagnostic system, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index
C1D21-05	Wheel module - System programming failures	- Incorrect wheels/tires installed (without tire pressure sensors - Winter wheels, spare wheel, etc) - Incorrect tire pressure sensor(s) installed - One or more tire pressure sensors damaged - One or more tire pressure sensor internal failures or signals corrupted by RF interference	- Check that the wheels and tires comply with the manufacturer's specification for the vehicle - Check that the correct tire pressure sensors are installed - Check the tire pressure sensors for damage. Install new tire pressure sensors as necessary - Using the manufacturer approved diagnostic system, check the snapshot data to determine the number of missing tire pressure sensor signals - If all tire pressure sensor signals are missing, check for sources of RF interference (electronic devices, chargers, non-genuine accessories, etc) - If not all tire pressure sensor signals are missing:, using the manufacturer approved diagnostic system, perform routine - Tire Pressure Monitoring - Tire Pressure Sensor Test. Install new tire pressure sensors as necessary
C1A56-85	Left Front Tire Pressure Sensor and Transmitter Assembly - Signal Above Allowable Range	Incorrect front left tire pressure sensor installed	NOTE: The DTC is set if a WU that is not capable to measure pressure values in the high range is fitted to the vehicle Using the manufacturer approved diagnostic system, perform routine - Tire Pressure Monitoring - Tire Pressure Sensor Test

			▪ If the fault persists, install a new front left tire pressure sensor
C1A58-85	Right Front Tire Pressure Sensor and Transmitter Assembly - Signal Above Allowable Range	▪ Incorrect front right tire pressure sensor installed	NOTE: The DTC is set if a WU that is not capable to measure pressure values in the high range is fitted to the vehicle ▪ Using the manufacturer approved diagnostic system, perform routine - Tire Pressure Monitoring - Tire Pressure Sensor Test ▪ If the fault persists, install a new front right tire pressure sensor
C1A60-85	Left Rear Tire Pressure Sensor and Transmitter Assembly - Signal Above Allowable Range	▪ Incorrect rear left tire pressure sensor installed	NOTE: The DTC is set if a WU that is not capable to measure pressure values in the high range is fitted to the vehicle ▪ Using the manufacturer approved diagnostic system, perform routine - Tire Pressure Monitoring - Tire Pressure Sensor Test ▪ If the fault persists, install a new rear left tire pressure sensor
C1A62-85	Right Rear Tire Pressure Sensor and Transmitter Assembly - Signal Above Allowable Range	▪ Incorrect rear right tire pressure sensor installed	NOTE: The DTC is set if a WU that is not capable to measure pressure values in the high range is fitted to the vehicle ▪ Using the manufacturer approved diagnostic system, perform

			routine - Tire Pressure Monitoring - Tire Pressure Sensor Test ▪ If the fault persists, install a new rear right tire pressure sensor
U0001-81	High Speed CAN Communication Bus - Invalid serial data received	▪ Invalid data received from another control module via the high speed CAN bus (powertrain)	▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-86	High Speed CAN Communication Bus - Signal invalid	▪ Invalid signal received from another control module via the high speed CAN bus (powertrain)	▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid signal source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-87	High Speed CAN Communication Bus - Missing message	▪ Missing message from another control module via the high speed CAN bus (powertrain)	▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-88	High Speed CAN Communication Bus - Bus off	▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0300-00	Internal Control Module Software Incompatibility - No sub type information	▪ Tire pressure monitoring system control module is not configured correctly	▪ Using the manufacturer approved diagnostic system, re-configure the tire pressure monitoring system control module with the latest level software

U2300-54	Central Configuration - Missing calibration	Car configuration file mismatch with vehicle specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary. Clear the DTCs and retest
U2300-55	Central Configuration - Not configured	Car configuration file mismatch with vehicle specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary. Clear the DTCs and retest
U2300-56	Central Configuration - Invalid/incomplete configuration	Car configuration file mismatch with vehicle specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary. Clear the DTCs and retest

U2300-64	Central Configuration - Signal plausibility failure	▪ Tire pressure monitoring system control module is not configured correctly ▪ Car configuration file mismatch with vehicle specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs ▪ Using the manufacturer approved diagnostic system, re-configure the tire pressure monitoring system control module with the latest level software ▪ Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary. Clear the DTCs and retest
U3000-51	Control Module - Not programmed	▪ Tire pressure monitoring system control module is not configured correctly	▪ Using the manufacturer approved diagnostic system, re-configure the tire pressure monitoring system control module with the latest level software